

Chapter 16 Homework - Singular Value Decomposition

Part B - Hand SVD

Compute $A^T A$:

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} = \boxed{\begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}}$$

Compute eigenvalues:

$$|\begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix} - \lambda I| = |\begin{bmatrix} 9-\lambda & 0 \\ 0 & 1-\lambda \end{bmatrix}| = (9-\lambda)(1-\lambda) = 0$$

$$\boxed{\lambda_1 = 9, \lambda_2 = 1}$$

Compute singular values:

$$\sigma_i = \sqrt{\lambda_i}$$

$$\sigma_1 = \sqrt{9}, \sigma_2 = \sqrt{1}$$

$$\boxed{\sigma_1 = 3, \sigma_2 = 1}$$

Σ matrix:

$$\boxed{\Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}}$$

A is the same as Σ because it is diagonal with positive entries, so

$$\boxed{U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, V^T = V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}$$